

Supplementary Information for

Multi-Image Encryption with Memristive Hopfield Neural Networks and Semi-Tensor Product

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Supplementary Tables:

Table S1: TestU01 test of three types of attractors

Battery	Number of tests	Sequence size	Parameters	Attractor 1	Attractor 2	Attractor 3	Result
				Suspicious P -values	Suspicious P -values	Suspicious P -values	
Rabbit	40	2^{30}	Standard	0	0	0	Pass
Alphabit	17	2^{25}	Standard	0	0	0	Pass
Block Alphabit	102	2^{25}	Standard	0	0	0	Pass

Table S2: PSNR and SSIM evaluation results

Images	PSNR(P-C)/dB	SSIM(P-C)	PSNR(P-D)/dB	SSIM(P-D)
KT	8.0863	0.0092	∞	1
GD	7.6383	0.0072	∞	1
CX	8.0812	0.0091	∞	1
AS	6.9178	0.0038	∞	1

Table S3: Encryption performance for pure black and pure white images

Images	Correlation			IE
	H	V	D	
Pure black	-0.00008	0.0006	0.0009	7.9994
Pure white	-0.0003	-0.00006	-0.0002	7.9994

Table S4: The encryption speed when encrypting images of different quantities and sizes

Image size	256×256	512×512	1024×1024
5 images (TET/AET(s))	0.0692/0.0139	0.2940/0.0588	1.2283/0.2456
10 images (TET/AET(s))	0.1116/0.0112	0.5831/0.0583	2.4438/0.2444
20 images (TET/AET(s))	0.1900/0.0095	1.1803/0.0590	4.7845/0.2392
40 images (TET/AET(s))	0.3598/0.0090	2.3402/0.0585	9.7320/0.2433